Sample Execution Traces

Order Matching Engine (Mock 2)

Note: these runs use `local_tester/driver.h`. `local_test9a/9b` replay them and check **both** stdout and the log file, character for character. Each run starts with an empty log file and order IDs starting at 1.

Scenario 1: Matching, priority, self-trade, settlement

Input	Output	Why
`REG alice 100000`	`OK`	
`REG bob 100000`	`OK`	
`REG carol 100000`	`OK`	
`SELL bob INFY 101 50`	`ORDER 1`	rests: asks 101:50
`SELL carol INFY 100 30`	`ORDER 2`	rests
`SELL bob INFY 100 40`	`ORDER 3`	rests behind #2 at 100 (later ID)

`BOOK INFY`	`ASK 101.00 50 (1)`
	`ASK 100.00 70 (2)`
	`---`	

`BUY alice INFY 101 100`	`ORDER 4`	sweeps #2, #3 (100) then 30 of #1 (101)
`BOOK INFY`	`ASK 101.00 20 (1)` `---`	#1 has 20 left
`BUY alice INFY 99 10`	`ORDER 5`	rests
`BUY carol INFY 99.5 5`	`ORDER 6`	rests
`BUY bob INFY 99.5 8`	`ORDER 7`	rests (bob)

`BOOK INFY`	`ASK 101.00 20 (1)`
	`---`
	`BID 99.50 13 (2)`
	`BID 99.00 10 (1)`
		two orders at 99.50

`SELL bob INFY 99 12`	`ORDER 8`	bob's own #7 skipped; hits #6 (5@99.50) then #5 (7@99)

`BOOK INFY`	`ASK 101.00 20 (1)`
	`---`
	`BID 99.50 8 (1)`
	`BID 99.00 3 (1)`
		#7 untouched, #5 has 3 left

`POS alice INFY`	`alice INFY 107 89277.00`	+30+40+30+7 shares; cash 100000−10030−693
`POS bob INFY`	`bob INFY -82 108220.50`	sold 40+30+12; cash +4000+3030+497.50+693
`POS carol INFY`	`carol INFY -25 102502.50`	sold 30, bought 5
`END`		

Log file after scenario 1

```
REGISTER alice 100000.00
REGISTER bob 100000.00
REGISTER carol 100000.00
ACCEPT #1 SELL bob INFY 50/50@101.00
ACCEPT #2 SELL carol INFY 30/30@100.00
ACCEPT #3 SELL bob INFY 40/40@100.00
ACCEPT #4 BUY alice INFY 100/100@101.00
TRADE INFY 30@100.00 B#4:alice S#2:carol
TRADE INFY 40@100.00 B#4:alice S#3:bob
TRADE INFY 30@101.00 B#4:alice S#1:bob
ACCEPT #5 BUY alice INFY 10/10@99.00
ACCEPT #6 BUY carol INFY 5/5@99.50
ACCEPT #7 BUY bob INFY 8/8@99.50
ACCEPT #8 SELL bob INFY 12/12@99.00
TRADE INFY 5@99.50 B#6:carol S#8:bob
TRADE INFY 7@99.00 B#5:alice S#8:bob
```

Scenario 2: Rejections, self-trade skip, cancels

Input	Output	Why
`REG alice 1000`	`OK`	
`REG alice 500`	`ERR Trader already registered`	duplicate: thrown, not logged
`REG bob 5000`	`OK`	
`REG eve -10`	`ERR Invalid cash`	negative cash
`BUY dave TCS 10 1`	`ORDER -1`	placeOrder never throws: logs REJECT, returns -1
`BUY alice TCS 100 20`	`ORDER -1`	$1000 < 100 \times 20$
`BUY alice TCS 50 0`	`ORDER -1`	Invalid quantity (no ID used)
`SELL bob TCS 0 5`	`ORDER -1`	Invalid price (no ID used)
`SELL alice TCS 40 10`	`ORDER 1`	first real order gets ID 1
`SELL bob TCS 45 10`	`ORDER 2`	
`BUY alice TCS 45 10`	`ORDER 3`	alice's own #1 (40, cheaper) skipped; trades 10@45 with #2
`BOOK TCS`	`ASK 40.00 10 (1)` `---`	only alice's #1 left
`CANCEL bob TCS 1`	`CANCEL FAIL`	#1 is alice's: Not your order
`CANCEL alice TCS 99`	`CANCEL FAIL`	Order not found
`CANCEL alice INFY 1`	`CANCEL FAIL`	unknown symbol: Order not found
`BOOK INFY`	`ERR Unknown symbol INFY`	getBook throws
`CANCEL alice TCS 1`	`CANCEL OK`	
`BOOK TCS`	`---`	empty book: just the separator
`POS alice TCS`	`alice TCS 10 550.00`	$1000 - 450$
`POS bob INFY`	`bob INFY 0 5450.00`	known trader, never traded INFY: 0
`POS zed TCS`	`ERR Unknown trader zed`	
`END`		

Log file after scenario 2

```
REGISTER alice 1000.00
REGISTER bob 5000.00
REJECT dave: Unknown trader dave
REJECT alice: Insufficient cash
REJECT alice: Invalid quantity
REJECT bob: Invalid price
ACCEPT #1 SELL alice TCS 10/10@40.00
ACCEPT #2 SELL bob TCS 10/10@45.00
ACCEPT #3 BUY alice TCS 10/10@45.00
TRADE TCS 10@45.00 B#3:alice S#2:bob
REJECT bob: Not your order
REJECT alice: Order not found
REJECT alice: Order not found
CANCEL #1 alice
```